

AI IT HELPDESK AGENT

Project Report

Architecture: Agent + RAG + Tools

Use Case: Diagnose common technical issues and recommend troubleshooting steps using a knowledge base.

Prepared as an academic project document

2. Project Overview

The project is an AI-powered first-line IT helpdesk agent designed to help users resolve everyday technical problems before they become formal support tickets.

The system combines an intelligent agent with Retrieval-Augmented Generation (RAG) and enabled tools. The agent understands the user's issue, retrieves relevant information from the knowledge base, and provides grounded troubleshooting guidance.

- Project title: AI IT Helpdesk Agent
- Primary role: First-line technical support
- Core architecture: Agent + RAG + Tools
- Knowledge sources are indexed for retrieval
- The system supports escalation when an issue needs human review

3. Problem Statement

Users frequently face common IT problems such as connectivity issues, software errors, account problems, or device configuration difficulties. Traditional support can require waiting for a technician or creating a ticket even when the issue is simple.

The project addresses this gap by providing an AI-based first-line support layer that uses the organization's available knowledge instead of relying only on generic answers.

- Reduce time spent on repetitive support questions
- Provide consistent troubleshooting guidance
- Use organization-specific knowledge
- Resolve incidents before they become tickets
- Escalate unresolved or sensitive issues for human review

4. Objectives & Proposed Solution

The objective is to build a grounded helpdesk assistant that can understand technical questions and guide users through practical troubleshooting steps.

The proposed solution is an Agent + RAG + Tools workflow. RAG supplies relevant knowledge, while the agent decides how that knowledge should be used to answer the user.

- Understand and classify the user's technical issue
- Retrieve relevant knowledge-base content
- Generate grounded troubleshooting responses
- Use available tools when an action is required
- Maintain a human-review path for unresolved incidents

5. Agentic AI Architecture

The project follows an agentic architecture in which the AI is responsible for interpreting the request, deciding the next step, using retrieval or tools, and producing a response.

Unlike a simple chatbot, the agent is connected to supporting capabilities and can follow a troubleshooting workflow.

- Input: user's technical issue
- Reasoning/decision layer: understand the issue and choose an action
- RAG layer: retrieve relevant knowledge
- Tools layer: perform supported actions
- Output: troubleshooting guidance or escalation

6. RAG – Retrieval-Augmented Generation

RAG improves the reliability of the helpdesk by retrieving relevant information from the indexed knowledge base before the final response is generated.

In the displayed project workspace, the Knowledge Base shows 5 indexed articles and the response mode is labelled as a grounded agent with RAG + tools enabled.

- Retrieve relevant articles or runbooks
- Use retrieved context as evidence for the response
- Reduce dependence on unsupported general knowledge
- Keep troubleshooting aligned with the team's documented knowledge
- Make responses easier to review and maintain

7. Tools Layer

Tools extend the agent beyond answering questions. In this project, the interface explicitly indicates that tools are enabled. The exact actions available depend on the configured implementation.

A typical IT helpdesk tool layer can support operations such as ticket handling, status checks, escalation, or other approved support actions.

- Tools are invoked when the workflow requires an action
- Tool usage can be controlled by the agent
- Actions should follow access and safety rules
- Unresolved cases can move toward human review
- Tool availability should be documented in the final implementation

8. Knowledge Base

The Knowledge Base is the main source of domain-specific troubleshooting information. Articles, FAQs, and runbooks can be indexed so the RAG layer can retrieve useful context.

The current workspace screenshot shows 5 knowledge sources/articles indexed and a dedicated Retrieval Layer section.

- Technical troubleshooting articles
- Frequently asked questions
- Step-by-step runbooks
- Known issue and resolution information
- Search by article content, tags, or runbooks

9. Working Process

The overall workflow begins when a user submits a technical issue. The agent interprets the request and determines whether it can be answered using the knowledge base or whether a tool/action is needed.

- 1. User describes the technical problem
- 2. Agent understands and classifies the issue
- 3. RAG retrieves relevant knowledge
- 4. Agent uses the retrieved context to formulate guidance
- 5. Tools are used when an action is supported
- 6. User follows troubleshooting steps
- 7. If unresolved, the issue can be escalated for human review

10. Key Features

The project interface and stated use case support several important features for an AI IT helpdesk.

- Grounded first-line agent
- Knowledge-base retrieval
- RAG + tools enabled response mode
- Knowledge source indexing
- New diagnosis workflow
- Escalation tracking
- Resolved-incident tracking
- Agent operational status and live context

11. Technologies / Tools Used

The implementation is presented as a web-based AI helpdesk workspace. The visible interface identifies the system as an AI IT Helpdesk and shows a grounded agent with RAG + tools enabled.

Because the exact backend stack is not fully visible in the supplied screenshot, this report avoids inventing specific programming frameworks or model names.

- AI agent layer
- Retrieval-Augmented Generation (RAG)
- Knowledge-base / document retrieval
- Tool-enabled agent workflow
- Web-based dashboard interface
- Human escalation / guardrail concept

12. User Interface / Dashboard

The dashboard provides a workspace for managing the knowledge base, monitoring incidents, and starting a new diagnosis. It also exposes the retrieval and agent context areas.

- Knowledge sources: 5 indexed articles shown in the supplied screenshot
- Open escalations: 0 at the time of the screenshot
- Resolved today: 0 at the time of the screenshot
- Response mode: Grounded agent
- RAG + tools: enabled
- Retrieval Layer: Knowledge Base
- Agent Context: Retrieval, Tools, Guardrail

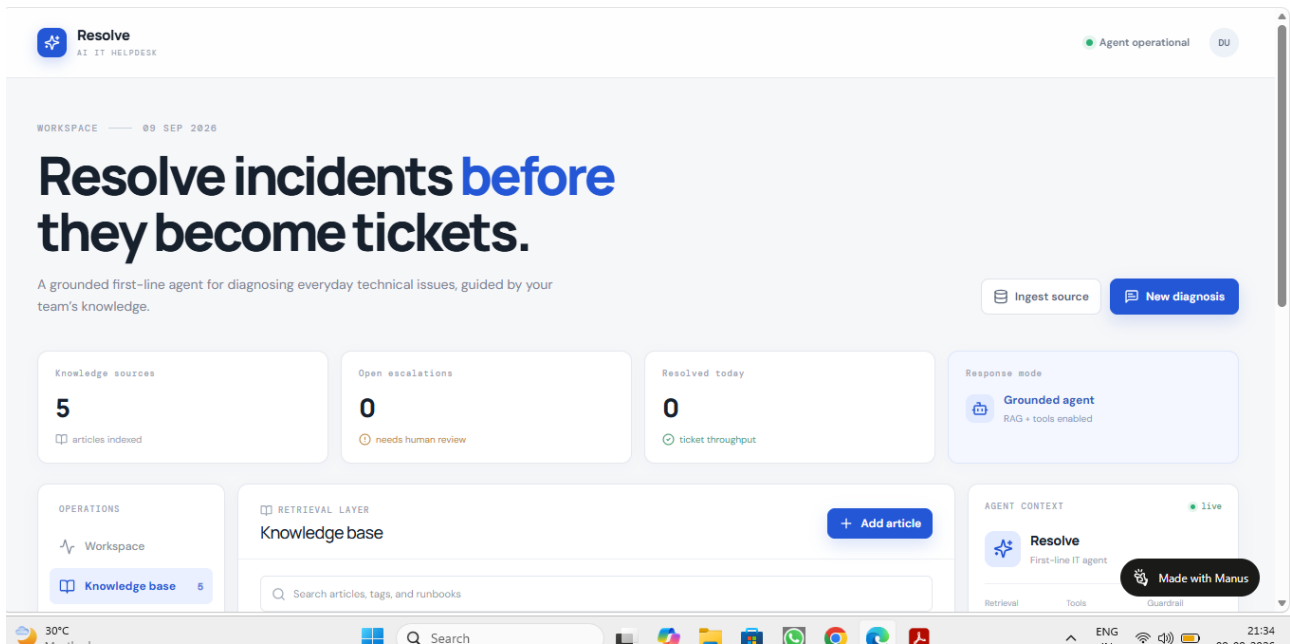


Figure: Supplied project dashboard screenshot.

13. Example Diagnosis Flow

Example scenario: a user reports that the college or office Wi-Fi is not connecting. The agent first interprets the issue, then retrieves relevant Wi-Fi troubleshooting content from the knowledge base.

The response can guide the user through safe, ordinary checks such as verifying the network, reconnecting, checking device settings, and following the organization's documented steps.

- User query → issue identification
- Retrieve Wi-Fi troubleshooting article
- Generate grounded step-by-step guidance
- Ask for the result of the troubleshooting
- Escalate if the documented steps do not resolve the issue

14. Agent + RAG + Tools: How They Work Together

The three components have different responsibilities but operate as one workflow.

- Agent: understands the request and controls the workflow
- RAG: finds the most relevant knowledge from indexed sources
- Tools: carry out supported actions or retrieve operational information
- Guardrails: help keep the workflow within approved boundaries
- Final response: clear, actionable, and grounded in available knowledge

15. Results Achieved

The supplied dashboard demonstrates that the project has an operational workspace with a grounded agent, indexed knowledge sources, retrieval, and tool support.

At the time of the supplied screenshot, the dashboard displayed 5 knowledge sources, 0 open escalations, and 0 resolved today. These are snapshot values and may change as the system is used.

- Grounded agent mode is visible
- RAG + tools are enabled
- Knowledge articles are indexed
- Retrieval layer is available
- Incident and escalation counters are present
- The interface supports starting a new diagnosis

16. Benefits of the Proposed Solution

An AI first-line helpdesk can improve the support experience by handling repetitive issues quickly while keeping the organization's knowledge at the center of the response.

- Faster first response for common issues
- Consistent troubleshooting instructions
- Reduced repetitive workload for IT staff
- Better use of existing documentation
- Potential reduction in unnecessary tickets
- Clear path to human escalation for difficult cases

17. Challenges Faced

Building a useful helpdesk agent requires more than connecting an AI model to a chat interface. The quality of the knowledge base, retrieval accuracy, tool permissions, and escalation logic all affect the final experience.

- Keeping knowledge-base articles accurate and updated
- Retrieving the correct article for ambiguous queries
- Preventing unsupported or overconfident answers
- Designing safe and reliable tool actions
- Handling issues that require human expertise
- Testing the agent across different user descriptions

18. Future Enhancements

The project can be extended into a more complete IT service management assistant.

- Add more knowledge articles and runbooks
- Improve retrieval with better tagging and ranking
- Add authenticated ticket creation and tracking
- Integrate approved IT service-management systems
- Add analytics for common incident categories
- Support multilingual user queries
- Add feedback loops to improve knowledge articles
- Improve escalation and human-agent handoff

19. Conclusion

The AI IT Helpdesk Agent demonstrates how Agentic AI, RAG, and tools can be combined to provide grounded first-line technical support.

The Agent handles the overall reasoning and interaction, RAG supplies organization-specific knowledge, and tools extend the system toward practical actions. Together, these capabilities can help resolve common incidents before they become tickets while preserving a human-review path for complex cases.

- Architecture: Agent + RAG + Tools
- Focus: everyday IT troubleshooting
- Knowledge-driven and grounded responses
- Action-oriented support workflow
- Scalable foundation for future IT automation

20. References

1. Project workspace supplied for this report: AI IT Helpdesk Agent web application.
 2. Supplied project dashboard screenshot used as implementation/output evidence.
 3. General concept reference: Retrieval-Augmented Generation (RAG) as a method for retrieving external knowledge before generation.
 4. General concept reference: Agentic AI systems that combine reasoning, retrieval, and tools to complete multi-step tasks.
- Note: Specific model names, programming frameworks, APIs, and tool names were not asserted where they were not visible in the supplied project material.